

PHIL HAYWARD

Senior Full-Stack Engineer &
Technical Lead

CONTACT

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AVAILABILITY

Right to Work
Full, unrestricted UK Right to Work

Mobility
Fully flexible for up to 60% UK-wide travel, including initial deployment in the West Midlands (Doncaster allows easy regional access)

CORE COMPETENCIES

Hands-on Engineering
C#, .NET 9 / .NET Core, ASP.NET Core, React, TypeScript, C++, Python, R, WPF

Systems & Integration
REST/gRPC API Design, Event-Driven Architecture (Azure Service Bus, SignalR), Microservices, Third-Party B2B/Supplier API Integrations

Cloud & DevOps
Azure (App Services, SQL, Key Vault, Container Apps), AWS, Docker, Bicep (IaC), PowerShell, CI/CD, OpenTelemetry

Forward-Deployed Skills
On-Site Requirements Gathering, Rapid Prototyping, Clinical Workflow Mapping, Live Product Demos, Remote User Training & Support

Healthcare & Regulated Domains
Clinical Trials Modelling, Anonymized Data Curation, Blinding/Privacy Protocols, GDPR/Consents, Zero Trust, RBAC

Leadership
Technical Mentorship, Cross-Functional Stakeholder Management, QA Function Building, Agile/Scrum

EDUCATION

BSc (Hons) Software Engineering (2:1)

PROFESSIONAL SUMMARY

Pragmatic, hands-on Senior Full-Stack Engineer and Technical Lead with 30 years of experience building secure, high-integrity applications in highly regulated environments. Specialising in **.NET / C# and Azure**, with deep domain expertise in the **healthcare, pharmaceutical, and clinical sectors** (13 years building clinical trials and biopharmaceutics data systems).

Operating at the intersection of rapid, production-ready coding and frontline stakeholder engagement, I excel at shadowing clinical/scientific end-users, identifying workflow bottlenecks, and building technical prototypes to solve immediate operational challenges. Adept at navigating ambiguity, leading complex system architectures, and translating on-the-ground insights into scalable core product features. Proven track record of managing high-stakes, cross-functional relationships and executing successful on-site deployments both UK-wide and internationally.

PROFESSIONAL EXPERIENCE

- Senior Full-Stack Engineer (Technical Lead)**

Dec 2024 – Present

Brand 'n Deliver — Sheffield (Hybrid)

Operating in a highly autonomous role, bridging the gap between core architecture and external B2B integrations to solve operational bottlenecks.

 - Systems Integration & Prototyping:** Integrated multiple complex third-party B2B supplier APIs (including XML product feeds with automated price auditing), extending the product offering and streamlining external data exchange for enterprise clients.
 - Hands-on Engineering:** Re-architected a visualisation platform from Azure Functions to .NET 8 containerised microservices. Defined API standards across a 470+ file codebase and implemented Azure Service Bus and SignalR for a highly responsive, event-driven user experience.
 - Rapid Deployment & DevOps:** Wrote the platform's entire Bicep IaC estate and PowerShell orchestrator, transitioning services from manual deployments to automated CI/CD pipelines to enable rapid, reliable feature releases.
 - Observability & Diagnostics:** Improved system reliability using Application Insights and OpenTelemetry. Diagnosed and resolved a critical production login failure for high-volume accounts by introducing a paged API, directly improving the end-user experience.
- Lead Software Engineer**

Nov 2021 – Oct 2024

Lockwood Publishing — Nottingham (Remote)

Led technical strategy, compliance, and cross-functional stakeholder management for a high-availability, global consumer platform handling 1 million monthly active users.

 - High-Stakes Stakeholder Management:** Led the implementation of a strict GDPR/Consents management system under pressing, revenue-impacting deadlines. Independently navigated high-stakes relationships across Legal, Marketing, and Client Services teams to ensure alignment and rapid adoption.
 - Solution Procurement & Integration:** Led the technical research and procurement of third-party consent APIs, managing the end-to-end integration and ensuring compliance in a highly regulated, ad-dependent free-to-play environment.
 - Leadership & Mentoring:** Led a 4-person cross-disciplinary engineering team, providing technical guidance, conducting peer code reviews, and mentoring a junior engineer to accelerate their career development.
 - Performance Engineering:** Led technical investigations and implemented performance enhancements for the TCP/IP/REST stack and cache strategies on AWS, ensuring optimal service reliability under heavy global load.

Sheffield Hallam University, 1996

Specialisation: C / Unix / Image Processing

CONTINUOUS LEARNING

Cloud & DevOps

Azure, AWS, Docker, GitOps, Linux System Administration

Modern Development

AI-Assisted Workflows (Copilot, Cursor, Claude), Agentic AI systems, Data Science

Senior Software Engineer

Certara (Simcyp) — Sheffield

Nov 2008 - Nov 2021

Deep healthcare/pharma domain role. Worked directly embedded with scientific and clinical end-users to build, demo, and support business-critical software.

- **On-Site Requirements Gathering & Frontline Collaboration:** Partnered directly with clinical stakeholders, including conducting on-site requirements gathering in Sweden, to map complex data gathering, blinding, and privacy requirements for an EU grant data-sharing system. Utilised rapid Excel-based prototyping and investigative user interviews to clarify vague requirements before engineering commenced.
- **Global Consortium Delivery:** Led the development of anonymized clinical trials data gathering and collaboration software used on a major project by a consortium of several of the world's top pharmaceutical companies, navigating complex data compliance and interoperability requirements.
- **Translating Field Insights to Core Product:** Identified a critical operational bottleneck where scientists spent excessive time on manual testing. Rapidly designed and deployed an automated output comparison and statistical variance tool for large ADME datasets (virtual population clinical trials), freeing up hundreds of hours of clinical staff time.
- **Rapid Prototyping:** Championed pragmatic problem-solving by prototyping complex systems of equations in R and Python for rapid functional verification, bypassing lengthy C++/C# production cycles to deliver immediate, verifiable value to the science teams.
- **End-User Training & Support:** Delivered comprehensive technical presentations, live product demonstrations, and remote tier-3 support to global scientific users, ensuring high adoption rates of newly deployed clinical trial modelling features.

Senior Software Engineer & Field Support

2001 - 2008

Synectics Systems — (Security Industry)

Customer-facing engineering and field support role across two distinct capacities, focused on high-profile, mission-critical command and control projects (e.g., Manchester City Centre, London's Ring of Steel, Rolls Royce).

- **UK-Wide Field Support & Development:** Travelled extensively across the UK to provide on-site field support, system diagnostics, and technical training for security operators, while simultaneously developing core software features and troubleshooting complex technical issues within challenging, legacy digital CCTV environments.
- **International Deployments:** Executed international technical engagements and on-site deployments in Canada and the USA, managing high-stakes stakeholder relationships and ensuring successful system handover and operational readiness for global clients.
- **Bid Support & Prototyping:** Actively supported the pre-sales bid process by developing technical prototypes and delivering live product demonstrations to senior, non-technical stakeholders to map workflow requirements.

PUBLICATIONS

Hayward P, et al. "Biopharmaceutics data management system for anonymised data sharing and curation: First application with OrBiTo IMI project." Comput Methods Programs Biomed. 2017;140:1–10. PMID: 28254086.